

Machine Learning: Tutorial Questions

Q1. Logistic regression is sometimes described as a *discriminative* classifier, while Gaussian Discriminant Analysis (GDA) is described as a *generative* classifier. Explain the distinction. Under what conditions might GDA outperform logistic regression, and when might logistic regression be preferred?

Q2. A Naïve Bayes model is trained on a text dataset in which the word “offer” never appears in any legitimate (non-spam) email. Describe exactly what goes wrong at test time when a legitimate email containing the word “offer” arrives, and explain how Laplace smoothing resolves the problem. Write the smoothed estimate $\hat{P}(\text{word} \mid y)$ explicitly.

Q3. Suppose you apply logistic regression to a two-class problem and plot the decision boundary. A colleague claims that “logistic regression always produces a straight-line decision boundary, so it cannot model non-linear class boundaries.” Is this claim correct? Explain, and describe one standard approach to overcome this limitation without changing the classifier itself.

Q4. Consider the following training set for a binary classification problem:

x_1	x_2	y
1	1	1
1	0	0
0	1	0
0	0	0

Using Naïve Bayes with binary features, compute $P(y = 1 \mid x_1 = 1, x_2 = 0)$. Show all intermediate probabilities.

Q5. A data point $\mathbf{x} = [4, 2]^\top$ is projected onto the first principal component $\mathbf{u}_1 = \frac{1}{\sqrt{5}}[2, 1]^\top$. Compute (a) the projection score $z = \mathbf{u}_1^\top \mathbf{x}$, (b) the reconstructed point $\hat{\mathbf{x}} = z \mathbf{u}_1$, and (c) the reconstruction error $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$.

Q6. Give two distinct failure modes of the K -means algorithm and, for each, suggest a concrete remedy.

Q7. Describe the elbow method for selecting K in K -means. What quantity is plotted against K , why does it always decrease as K increases, and what does the elbow indicate?

Q8. A binary classifier on a heavily imbalanced dataset (95% negative, 5% positive) achieves 95% accuracy by predicting *negative* for every example. Explain why accuracy is a misleading metric here. Name two better metrics and give their formulas.

Q9. Consider a dataset in $\mathbb{R}^{200}$ (200 features, 1000 examples). The eigenvalues of the covariance matrix, in decreasing order, are $\lambda_1 = 60$, $\lambda_2 = 40$, $\lambda_3 = 25$, $\lambda_4 = 15$, with the remaining 196 eigenvalues summing to 60.

- (a) What is the total variance in the dataset?
- (b) What fraction of the variance is explained by the first three principal components?
- (c) How many components are needed to explain at least 70% of the variance?

Q10. Dataset: $\{3, 5, 11, 13\}$, $K = 2$. Initial centroids: $\mu_1 = 4$, $\mu_2 = 12$. Carry out one complete iteration of K -means (assignment then update). State whether the algorithm has converged.

Q11. A decision tree node contains 12 training examples: 7 of class A and 5 of class B. A candidate split produces a left child with 6 examples (all class A) and a right child with 6 examples (1 class A, 5 class B).

- (a) Compute the Gini impurity of the parent node.
- (b) Compute the weighted Gini impurity after the split.
- (c) Compute the reduction in impurity and state whether this is a good split.

Q12. A decision tree achieves 100% training accuracy but 60% test accuracy. Name this phenomenon and describe two strategies—one that operates *before* the tree is fully grown and one that operates *after*—to mitigate it.

Q13. In Gaussian Discriminant Analysis with a shared covariance matrix Σ , the parameters ϕ , μ_0 , μ_1 , and Σ are estimated by maximum likelihood. Write down the closed-form MLE for each of the four quantities (you do not need to derive them, just state the formulas).

Q14. Two classifiers are compared on the same binary classification task. Classifier A has precision 0.80 and recall 0.60. Classifier B has precision 0.65 and recall 0.85. (a) Compute the F1 score of each. (b) In a medical diagnosis application where missing a positive case is very costly, which classifier is preferable, and why?

Q15. State the gradient descent update rule for logistic regression (without regularisation). Define every symbol you use. Explain why the cross-entropy loss is preferred over squared error for training a logistic regression model.

Q16. Suppose you apply PCA to a dataset and retain the top k components. Describe two distinct criteria a practitioner might use to choose k , and give one practical consideration that might lead you to choose a *smaller* k than the criteria suggest.

Q17. Describe the relationship between GDA and logistic regression. Specifically: if the GDA modelling assumptions hold exactly, which classifier achieves lower asymptotic error, and why? What happens when the assumptions do not hold?

Q18. A dataset has $n = 6$ points. A bootstrap sample of size 6 is drawn with replacement.

- (a) What is the probability that a specific point $x^{(i)}$ is *not* included in the bootstrap sample?
- (b) Approximately what fraction of the original training data is excluded on average? What are these excluded points called, and how are they used?

Q19. A regression tree is to be fit to the following five points:

$$(x, y) \in \{(1, 3), (2, 5), (3, 4), (8, 9), (9, 11)\}.$$

Consider a split at $x = 5$ (left: $x \leq 5$, right: $x > 5$). Compute the weighted sum of squared errors (SSE) after this split and compare it to the SSE before splitting (treating all five points as one node).

Q20. The Naïve Bayes assumption states that features are conditionally independent given the class label. Give a realistic example where this assumption is clearly violated. Despite the violated assumption, Naïve Bayes often performs surprisingly well in practice. Provide one reason why.